Regional EMU Fleet Mid-Life Overhaul (18 trainsets)

Project owner	Rolling Stock Engineering	**Budget (CAPEX)**	€ 14,500,000
Planned start	Q1 2027	**Planned completion**	Q4 2028

1. Project description

Mid-life overhaul of 18 four-car regional electric multiple units (72 cars, 144 bogies) after 15 years in service. Scope covers complete running-gear refurbishment, brake system renewal, passenger door replacement and HVAC upgrade to low-GWP refrigerant. Each trainset is out of service for 10 weeks; a maximum of 3 units may be in the workshop at any time to protect timetable availability.

2. Objectives

- Extend fleet service life by 15 years
- Reduce brake-related failures by 40%
- Meet PRM TSI accessibility requirements
- Comply with EU F-gas regulation (R134a phase-out)

3. Bill of materials

#	Item ID	Description	Standard	Qty	Supplier ID	Supplier
01	014-001	Bogie frame, welded, motor bogie	EN 15085-2 CL1	36 pc	SUPP_09733	Advanced Systems 379
02	014-002	Bogie frame, welded, trailer bogie	EN 15085-2 CL1	108 pc	SUPP_00410	Advanced Technologies 680
03	014-003	Monobloc wheel Ø920 mm, ER7	EN 13262	576 pc	SUPP_04507	Precision Logistics 645
04	014-004	Axlebox bearing unit TBU 130	EN 12082	576 pc	SUPP_04013	Global Manufacturing 285
05	014-005	Brake caliper unit, pneumatic, with parking brake	EN 15827	288 pc	SUPP_03658	Global Logistics 816
06	014-006	Axle-mounted brake disc Ø640 mm	EN 14535-1	576 pc	SUPP_02287	Dynamic Industries 204
07	014-007	Sintered brake pad set	UIC 541-3	1,152 set	SUPP_01680	Prime Manufacturing 114
08	014-008	Brake control unit (EP) incl. wheel-slide protection	EN 15734	72 pc	SUPP_08936	Elite Logistics 278
09	014-009	Secondary air spring with emergency rubber	EN 13597	288 pc	SUPP_01425	Advanced Solutions 335
10	014-010	Yaw damper, hydraulic	EN 13802	288 pc	SUPP_09675	Precision Group 666
11	014-011	Electric plug-sliding passenger door, 1300 mm	EN 14752	288 pc	SUPP_06913	Global Technologies 270
12	014-012	Saloon HVAC unit 34 kW, R513A	EN 14750-1	72 pc	SUPP_00521	Advanced Technologies 773

4. Procurement notes

- Suppliers listed are the current preferred source per item; alternatives may be proposed after risk review.
- All items require ISO/TS 22163 (IRIS) or equivalent certification and first article inspection before series delivery.
- Supplier due diligence per LkSG / CSDDD is required before purchase order release.

SIMULATED DEMO DOCUMENT – fictitious project for hackathon use. Supplier IDs and names refer to supplier_risk_dataset.csv.